</HTML>

ULTIMATE

HTML NOTEBOOK

By CodeWithSouma

HTML Notes

The word **hypertext markup language** is composed of the word's "**hypertext**" and "**markup language**". The term "**hypertext**" refers to the linking of text with other documents and "**markup language**" refers to a language that uses a set of tags.

PREFACE

Welcome to the "Ultimate HTML Handbook," your comprehensive guide to mastering HTML programming. This handbook is designed for beginners and anyone looking to strengthen their foundational knowledge of HTML, the fundamental language for creating web pages.

PURPOSE AND AUDIENCE

This handbook aims to make web development accessible and enjoyable for everyone. Whether you're a student new to coding, a professional seeking to enhance your skills, or an enthusiast exploring HTML, this handbook will definitely be helpful. HTML's simplicity and universal use make it an essential starting point for anyone interested in web development.

STRUCTURE AND CONTENT

The handbook is divided into clear, concise sections, each focused on a specific aspect of HTML:

- Fundamentals of HTML: Learn the basic structure and elements of HTML.
- Styling with CSS: Understand how to enhance your HTML pages with CSS.
- Interactive Elements: Explore how to create dynamic and interactive content using HTML.
- Advanced Topics: Dive deeper into more complex HTML techniques and best practices.

WHY HTML?

HTML is the backbone of the World Wide Web, providing the structure and semantics for web pages. It is essential for anyone looking to build a career in web development or simply understand how websites work.

HTML's versatility and ease of use make it an ideal starting point for aspiring web developers.

ACKNOWLEDGEMENTS

I extend my gratitude to all the educators, developers, and contributors whose insights and expertise have shaped the content of this handbook. Special thanks to the web development community for their continuous support and inspiration.

CONCLUSION

Learning web development through HTML can be both rewarding and empowering. The "Ultimate HTML Handbook" aims to make your journey into web development smooth and enjoyable. Use this guide as your companion to master HTML and creating impactful web experiences.

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CHAPTER 0 – INTRODUCTION

HTML, which stand for "Hyper Text Markup Language," is the foundational language of the web. It's used to create and structure websites. By utilizing HTML tags, we can define the appearance and layout of a website. With a good grasp of these tags and their proper usage, creating beautiful websites becomes straightforward and efficient!

THEN WHY CSS & JAVASCRIPT

HTML is used to define the layout of a page, providing a barebone structure for the content.

CSS is used to add styling to that barebone page created using HTML.

JavaScript is used to program logic for the page layout e.g. What happens when a user hovers on a text, when to hide or show elements etc.

A BEAUTIFUL ANALOGY

- HTML = Car body (only metal)
- CSS = Car paint, decoration etc.
- JavaScript = Car engine + Interior logic

We will start learning how to build beautiful websites in this course.

INSTALLING VS CODE

We can use any text editor of our choice. Here I am using VS Code because it is lightweight, opensource & from Microsoft.

Note: You can write HTML even in Notepad. Text editors like VS Code just makes these things easier.

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CHAPTER 1 – CREATING OUR FIRST WEBSITE

We start building a website by creating a file named index.html. index.html is a special filename which is presented when the website root address is typed.

A BA ASIC HTML PAGE

```
<!Doctype html> <!-- Specifies this is an html 5 document -->
<html> <!-- Root of an HTML page -->
<head> <!--  Contains page metadata -->
        <title>Soumaprit's Website</title> <!--  Contains title -->
</head>
<body> <!--  The main body of the page (rendered by the browser) -->
        <h1> <!--  This is a heading </h1> - heading tag -->
        <b>My paragraph </b> <!--  Paragraph tag -->
</body> <!--  Closing body tag -->
</html>
```

A tag is like a container for either content or other HTML tags.

IMPORTANT NOTES

- <head> & <body> tags are children of HTML tag.
- HTML is the parent of <head> & <body> tags.
- Most of the HTML element have opening & closing tag with contact in between opening & closing tags.
- Some HTML tags have no content. These are called Empty elements e.g.
- We can either use .htm or .html extension.
- You can use "inspect element "or "view page source "option from Chrom to look into a website's HTML Code.

COMMENTS IN HTML

Comments in HTML are used to mark text which should not be parsed. They can help document the source code.

```
<!--  HTML COMMENT  -->
```

CASE SENSITIVITY

HTML is a case insensitive language . <H1> and <h1> tags are the same.

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CHAPTER 1 – PRACTICE SET

1. Inspect your favourite website and change something on the page which is displayed.
2. Go to your favourite website and try to view the page source and write the exact lines of code. Does it clone the website? Why?
3. Write any HTML code inside a text file. Does it work if you write it using notepad?

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CHAPTER 2 – BASIC HTML TAGS

We can add elements inside the body tag to define the page layout.

HTML ELEMENT An HTML

element consists of everything from the starting tag to the ending tag.

```
<body> <!-- Opening tag -->
        Content
</body> <!-- Closing tag -->
```

HTML ATTRIBUTES

HTML attributes are used to add more information corresponding to an HTML tag. ***Example:***

```
<a href="https://soumaprit10.github.io/SoumapritPaul/#"/> Soumaprit </a>
```

We can either use single or double quotes in attributes.

THE HEADING TAG

Heading tag is used to mark headings in HTML. From h1 to h6, we have tags for the most important to the least important heading.

```
<h1> Most important heading </h1>
<h2> Another heading H2 </h2>
<h3> Another heading H3 </h3>
<h4> Another heading H4 </h4>
<h5> Another heading H5 </h5>
<h6> Another heading H6 </h6>
```

Note: We should not use HTML headings to make text thick or bold.

THE PARAGRAPH TAG

Paragraph tags are used to add paragraph to an HTML page.

```
<p> This is a paragraph </P>
```

THE ANCHOR TAG

The Anchor Tag is used to add links to an existing content inside an HTML page.

```
<a href="https://google.com"> click me </a>
```

THE IMG TAG

<img>tag is used to add images in an HTML page.

```
<img src="image.jpg">
```

BOLD, ITALIC AND UNDERLINE TAGS

We can use **bold**, *italic* and <u>underline</u> tags to highlight the text as follows:

```
<b> This is bold </b>
<i> This is italic </i>
<u> This is underline </u>
```

BR TAG

The <br> tag is used to create line breaks in an HTML document.

```
<br>
```

BIG AND SMALL TAGS

We can make text a bit larger and a bit smaller using and tags respectively.

```
<big> Hello world </big>
<small> Hello world </small>
```

HR TAG

<hr> tag in HTML is used to create a horizontal ruler often used to separate the content.

```
<hr>
```

SUBSCRIPT & SUPERSCRIPT

We can add subscript and superscripts in HTML as follows:

```
This <sub> is </sub> subscript.
This <sup> is </sup> superscript.
```

PRE TAG

HTML always ignores extra spaces and newlines. In order to display a piece of text as is, we use pre tag.

```
<pre>
    This is written.
      using pre
         tag
</pre>
```

CHAPTER 2 – PRACTICE SET

1. Create an HTML page with a heading (title heading), a primary heading and a subheading. Which tags did you use?

2. Create a page with 5 wallpaper images taken from the internet.

3. Use <br> and <hr> tags to display a piece of text with line breaks.

4. Try to write the following Chemical equation using HTML.

   C + O2 = CO2

5. Try to write a Wikipedia article using HTML.

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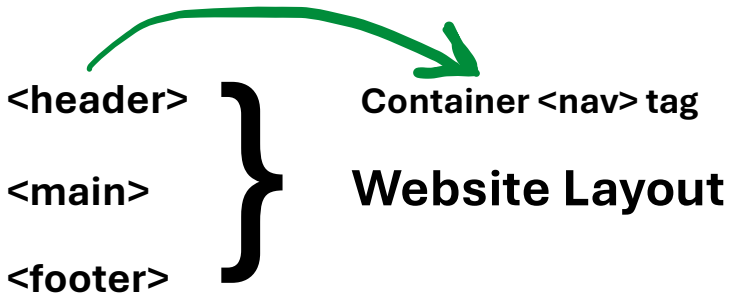
When we use the right tag in right place, it results in a better page layout, better indexing by search engines and better user experience.

We use the following tags to get the job done.



Inside the main tag insert the following tags:

```
<!-- The main opening tag -->
<main>
  <!-- A page section -->
  <section>
    <!-- A self-contained content -->
    <article>
      <!-- Content goes here -->
    </article>
  </section>
  <!-- Content aside from the main content (e.g. Ads, sidebars) -->
  <aside>
    <!-- Aside content goes here -->
  </aside>
</main>
<!-- The main closing tag -->
```

Creating a page like this is not necessary but it creates a readable & structured layout. Also, they are useful for SEO.

LINK ATTRIBUTES

```
<!-- Contact page opens in the same tab -->
<a href="/contact">Contact</a>
<!-- Contact page opens in a new tab -->
<a href="/contact" target="_blank">Contact us</a>
```

We can put any content inside an anchor tag (images, headings etc are all allowed).

If the page is inside a directory, we need to make sure that we link to the correct page (Same applies to img tag as well.)

We can add links to images like this.

```
<a href="/about"> <img src="a.jpg"  width="120"> </a>
```

THE DIV TAG

The tag is often used as a container for other elements. It is a block-level element, meaning it always takes up the full width available.

```
<div>
    <h1> This is a heading inside a div. </h1>
    <p> This is a paragraph inside a div. </p>
</div>
```

THE SPAN TAG

The tag is an inline container, meaning it only takes up as much width as necessary.

```
<p> This is a <span>highlighted </span> word in a sentence. </p>
```

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CHAPTER 3 – PRACTICE SET

1. Create an SEO friendly website using HTML.

2. Create an HTML page which opens google when clicked on an image.

3. Create a website which has your 5 top used websites bookmarked. The links should open in a new tab.

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CHAPTER 4 – LISTS, TABLES & FORMS

LISTS

Lists are used to display content which represents a list.

UNORDERED LIST

An unordered list is used to list items that do not have a specific order.

```
<ul>
    <li>Home</li>
    <li>About</li>
</ul>
```

ORDERED LIST

An ordered list is used to list items in a specific order, typically numbered.

```
<ol>
    <li>Phone</li>
    <li>PC</li>
    <li>Laptop</li>
</ol>
```

TABLES

The <table> tag in HTML is used to define tables, which are used to format and display tabular data.

- tag: Used to display table row.
- tag: Used to display table data.
- tag: Used in place of table data for displaying table headers.

We can define as many table rows as we want. To add a caption to the table, we use <capital> tag inside table.

- <thead> tag: Used to wrap table head (caption & <tr> with <th>)
- <tbody> tag: Used to wrap the table body.

Example:

```
<table>
   <caption>Students Report</caption>
   <thead>
      <tr>
         <th>Name</th>
         <th>Grade</th>
      </tr>
      <tr>
         <td>Rohan</td>
         <td>A++</td>
      </tr>
      <tr>
         <td>Soumaprit</td>
         <td>D</td>
      </tr>
   <tbody>
</table>
```

COLSPAN ATTRIBUTE

This attribute is used to create cells spanning multiple columns.

```
<!--   span 3 columns  -->
<th colspan="3">Harry</th>
```

HTML FORMS An HTML

tag is used to create a form that collects input from users.

```
<from>
   <!--   Elements of form  --->
</from>
```

There are different form elements for different kinds of user input.

- **<input> element:** Can be of type text, checkbox, radio, button and submit. We also have a 'file' type.
- **<textarea> element:** Defines a multiline text input 'cols' and 'rows' attributes can be used to size the text area.
- **<select> element:** Defines a drop-down list.

Note: you don't have to remember all the tags, you will automatically memorize them with practice.

EMBEDDING VIDEOS

To embed videos in HTML, you can use the tag along with various attributes to control its behaviour.

```
<video src =”soumaprit.mp4”>ERROR</video>
```

ATTRIBUTES FOR VIDEO

We can use the following attributes:

- **src:** Specifies the URL of the video file (harry.mp4 in this case).
- **width:** Adjusts the width of the video player. Height adjusts automatically to maintain aspect ratio.
- **controls:** Displays video controls such as play, pause, volume, etc.
- **autoplay:** Automatically starts playing the video when the page loads.
- **loop:** Causes the video to automatically start over from the beginning when it reaches the end.
- **preload:** Specifies whether the video should be loaded when the page loads (auto, metadata, none).

```
<video width="640" autoplay loop>
   <source src="harry.mp4" type="video/mp4">
   Your browser does not support the video tag.
</video>
```

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CHAPTER 4 – PRACTICE SET

1. Create an HTML page with video embedded inside it.
2. Replace the video in question no 1 with a YouTube video.
3. Create an HTML form for a travel website to book a vacation.
4. Create a table displaying score of cricket Players in a match using HTML.

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CHAPTER 5 – SEO

We will focus only on HTML standpoint of SEO. We will not be looking into keyword building and content optimization aspect of SEO.

TYPES OF SEO

- On page SEO (This can be done by HTML developers).
- Off page SEO.

HTML SEO

HTML developers can implement SEO using the following techniques:

1. **Title Tag:** Set a clear and descriptive <title> tag that accurately reflects the content of the page.
2. **Meta Description:** Provide a concise summary of the page content using the <meta> tag.
3. **URL Slug:** Use a clean and readable URL structure that includes relevant keywords.
4. **Meta Author Tag:** Optionally include the author information in a <meta> tag
5. **Favicon:** Use a favicon to enhance brand recognition and usability.
6. **Image Optimization:** Compress images to improve page load times and include descriptive alt attributes.
7. **Optimize Resources:** Remove unused HTML, CSS, and JavaScript files, and minify/compress them to reduce page load times.
8. **Semantic HTML:** Use appropriate HTML tags (<header>, <nav>, <article>, <footer>, etc.) to structure the content logically, which can improve SEO indirectly by making the content more accessible and understandable to search engines.

```
<head>

   <title> Example Page – SEO Best Practices </title>

   <meta name="description" content="this is an example  page demon string
SEO best practices in HTML." >

   <meta name="author" content="Soumaprit">

   <link rel="icon" href="favicon.ico" type="image/x-icon">

</head>
```

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Basic HTML Structure

What is the Basic Structure of an HTML Page?

Every HTML document follows a basic structure. This structure tells the browser how to read and display the content.

Template of a Basic HTML Page:

```
<!DOCTYPE html>
<html>
   <head>
      <title>Page Title</title>
   </head>
<body>
   <!-- Your content goes here -->
</body>
</html>
```

Explanation of each part:

1) <!DOCTYPE html>
   - Declares that this is and HTML5 document.
   - Must be the line in the file.
2) <html>...</html>
   - The root element of the page.
   - Wrape all the content of your HTML document.
3) <head>...</head>
   - Content meta-information about the page.
   - This can include:
   - The page <title>
   - Link to CSS file
   - Meta tags
4) <title>...</title>
   - Sets the name show on the browser tab.
5) <body>...</body>
   - Contains everything visible on the page.
   - You'll place text, links, images, forms, etc. here.

Headings and Paragraphs:

Headings in HTML

HTML provide 6 levels of heading:

<h1> -Biggest

<h2> - Subheading

<h3> - Smaller subheading

<h4> , <h5>, <h6> - Even smaller headings

Paragraphs in HTML :

Paragraphs are written using the <p> tag.

Line Breaks:

if you want to break a line without starting a new paragraph, use the <br> tag.

Bold Text:

Use the <b> or tag to make text blod.

Italic Text:

Use the <i> or tag to italicize text.

 gives extra emphasis and has meaning, especially for accessibility.

Underline Text:

Use the <u> tag to underline text.

Strikethrough Text

Use the <s> or <del > tag to show deleted or crossed-out text.

Superscript and Subscript

Use <sup> for superscript (above line), <sub> for subscript (below line).

HTML Comments

Comments are notes in your HTML code that are ignored by the browser. They are useful for explaining code or leaving reminders.

<!-- This is a Comment -->

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Creating Links in HTML:

HTML uses the The **<a>** tag to create links. **href** attribute tells the browser where the link should go.

< a href=”https://www.exmple.com”>Visite Example</a>

Opening Links in a New Tab

< a href=”https://www.exmple.com” target=”_blank”>Visite Example</a>

Linking to Other Pages (Internal links):

< a href=”index.html”>Visite Example</a>

Link to a Section on the Same Page

< a href=”#contact”>Visite Example</a>

……

<h2 id=”contact”>Contact Section </h2>

Email and Phone Links:

Email Link:

<a href="mailto:hello@example.com">Send Email</a>

Phone Link:

<a href=”tel:+1234567890”>Call us </a>

Images in HTML

<img src="image.jpg" alt="Description of image">

Image Size: Width and Height

<img src="pic.jpg" alt="Sample" width="200" height="150">

Lists in HTML

Unordered List – Bulleted list

Ordered List – Numbered list

Description List – List of terms and their descriptions

Unordered List

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<ul>

<li>Soumaprit</li>

<li>Coding</li>

</ul>

Ordered List

<ol>

<li>Soumaprit</li>

<li>Coding</li>

</ol>

Tables in HTML

Creating Tables

Use the <table> tag to create table . Inside a table, use:

- <tr> for table rows
- <td> for table data (cells)
- <th> for table headers

Thank You ☺

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